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Smoothness as a Constraint for Stable Humanoid Locomotion

DeCap Decouples Upper- and Lower-Body Smoothness into Separate Constraint Groups, Cutting Real-Robot Action Rate by 2.50× Without Reward Retuning Across Terrains

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Reward-based smoothness penalties compete with locomotion objectives and treat the body uniformly. DeCap reformulates whole-body smoothness as explicit upper- and lower-body constraints with a bounded barrier penalty, reducing upper-body action rate by 2.50× and acceleration by 2.18× on real humanoids while transferring to new terrains without reward retuning.

AT A GLANCE

Problem: Reward-based smoothness competes with task rewards and treats upper and lower body uniformly, ignoring their different physical roles in humanoid stability.

Why it matters: Jerky upper-body motions destabilize center-of-mass trajectories; over-smoothing the lower body prevents reactive stepping on uneven terrain.

Method: DeCap: separate tight/loose bounded barrier constraints for upper/lower body, formulated as an RL constraint term.

Result: 2.50× upper-body action-rate reduction, 2.18× acceleration reduction; terrain transfer without reward retuning.

Evidence: Real humanoid whole-body control evaluation on flat ground, stairs, slopes, and stepping stones; 4-axis ablation study.

The Big Picture

Humanoid robots are expected to walk, carry objects, and navigate complex terrains while remaining stable and human-like in motion. Reinforcement learning has enabled impressive locomotion policies, but the physical quality of the motion—smoothness, absence of oscillations, predictability—has been an afterthought, typically handled by adding penalty terms to the reward function. This paper argues that treating smoothness as a reward

penalty is the wrong abstraction for two reasons:

First, it creates objective competition. A smoothness penalty competes with speed rewards and stability bonuses during gradient descent. The resulting trade-off must be re-tuned for every new terrain, speed target, and robot variant—a laborious process that does not transfer.

Second, it treats the body uniformly. The upper body (torso, arms) must be tightly regulated to prevent tipping—inertial forces from fast arm motion directly perturb the center-of-mass. The lower body (hips, knees, ankles) must remain reactive to ground-contact perturbations; over-smoothing the legs leads to falls on uneven terrain.

DeCap (Decoupled Constraint-aware Policy) addresses both problems by reformulating smoothness as explicit, body-part-specific constraints within the RL objective.

Why Existing Approaches Fall Short

Reward-based action-rate penalties are soft: the optimizer can trade smoothness for task performance whenever doing so increases total reward. The penalty coefficient requires extensive per-terrain tuning.

Lipschitz-constrained policies bound the policy’s output sensitivity globally but cannot differentiate between upper and lower body requirements.

Model predictive control (MPC) provides smooth trajectories by design but requires accurate dynamics models and fails on highly uncertain terrain contacts.

Trajectory imitation from motion capture data inherits human smoothness but is brittle to out-of-distribution perturbations and terrain types.

Core Mechanism

Decoupled Constraint Groups. DeCap partitions joints into upper-body set $\mathcal{U}$ (torso, arms, neck) and lower-body set $\mathcal{L}$ (hips, knees, ankles), reflecting their distinct physical roles. Smoothness is measured as joint action rate:

$$\dot{a}_k^{(j)} = \frac{a_k^{(j)} - a_{k-1}^{(j)}}{\Delta t}$$

Two separate constraint limits are set: tight κ_U for the upper body (derived from human motion capture statistics), loose κ_L for the lower body (set to allow reactive stepping).

Bounded Barrier Penalty. Instead of a standard log-barrier (which diverges at the constraint limit and causes training instability), DeCap uses a bounded barrier that activates proactively in the range $[\alpha\kappa, \kappa)$ and saturates at

the limit:

$$\phi(c, \kappa) = \begin{cases} 0 & c \leq \alpha\kappa \\ \frac{(c - \alpha\kappa)^2}{\kappa - \alpha\kappa} & \alpha\kappa < c < \kappa \\ \kappa - \alpha\kappa & c \geq \kappa \end{cases}$$

where $c = \|\dot{\mathbf{a}}_k\|_2$ is the action-rate norm and $\alpha \in (0, 1)$ is the activation threshold fraction.

Technical Formulation

Upper-body constraint barrier reward term:

$$R_{\text{smooth}}^U = -\lambda_U \cdot \phi(\|\dot{\mathbf{a}}_k^U\|_2, \kappa_U)$$

Lower-body constraint barrier:

$$R_{\text{smooth}}^L = -\lambda_L \cdot \phi(\|\dot{\mathbf{a}}_k^L\|_2, \kappa_L)$$

Second-order (acceleration) constraint for the upper body:

$$\ddot{a}_k^{(j)} = \frac{\dot{a}_k^{(j)} - \dot{a}_{k-1}^{(j)}}{\Delta t}, \quad R_{\text{acc}}^U = -\lambda_{\text{acc}} \cdot \phi(\|\ddot{\mathbf{a}}_k^U\|_2, \kappa_U^{\text{acc}})$$

Full RL objective:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{task}}(\theta) + R_{\text{smooth}}^U + R_{\text{smooth}}^L + R_{\text{acc}}^U$$

Learning or Inference Procedure

1. Partition robot URDF joints into $\mathcal{U}$ and $\mathcal{L}$.
2. Set κ_U from human motion-capture action-rate statistics; set κ_L empirically to allow reactive stepping.
3. Train PPO in Isaac Gym on flat ground with domain randomisation.
4. Deploy on real humanoid; evaluate on flat ground and 3 novel terrains (stairs, slopes, stepping stones) without reward retuning.

What the Guarantee Says

The bounded barrier provides a formal advantage over log-barriers: the gradient is bounded everywhere (including at the constraint limit), preventing catastrophic gradient magnitudes that destabilise RL training. For $c \geq \kappa$, the barrier saturates at $\kappa - \alpha\kappa$, a finite value that continues to penalise violations while remaining numerically stable. The proactive activation at $\alpha\kappa$ teaches the policy to stay away from the limit before it is reached.

Experimental Findings

- **Upper-body action rate:** 2.50× reduction over reward-based smoothness baseline on flat ground

- **Upper-body acceleration:** 2.18× reduction
- **Lower-body smoothness:** Marginal improvement; reactive range preserved
- **Transient motion:** Reduced oscillations in torso and arm joints during gait transitions
- **Terrain transfer:** Fixed constraints transfer to stairs, slopes, and stepping stones without reward retuning; reward-based baselines require significant coefficient adjustment on each terrain

Ablations and Interpretation

No decoupling (single constraint group): Upper-body smoothness improves, but lower-body reactivity degrades on uneven terrain, leading to falls; confirms that separate limits are necessary.

Standard log-barrier: Training instability due to gradient explosions near the constraint limit; DeCap’s bounded barrier resolves this.

Reward term instead of constraint: Smoothness improvement is $\approx 40\%$ of DeCap’s at matched task-reward coefficient, confirming that the constraint formulation provides direct, stable control over the smoothness quantities.

α sensitivity: $\alpha = 0.7$ provides the best balance; smaller α (earlier activation) over-constrains and reduces task reward by $\approx 5\%$.

Reference

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<https://arxiv.org/abs/2609.24552>